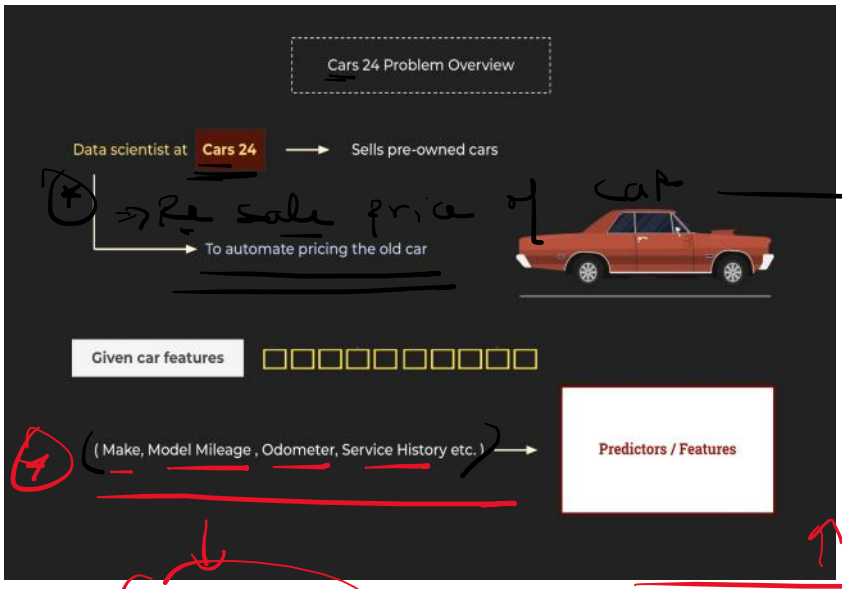
LINEAR REGRESSION

Agenda →

- Cars 24 Data
- Linear Reg
- intuition of LR
- maths → Algebraic
- code → sklearn, scratch

4 Classes

- Regression → continuous
- Classification → 0 1



CARS 24 →

target variable =

DATASET → Resale price of the car

6x Rules V/s ML algos

i/p → ML algo → pred

1 feature → ML model

(*) 1 feature ⇒ ML model

↳ target

Features of car

goal → Resale price (target) of that car

↳ Predictor vars → (*) model, mileage ...



ML model → Price of car

→ i/p

→ independent var

o/p or dependent var

→ Regression (continuous) → supervised → (*) unsupervised

issues → (1) Not Scaled

(2) Categorical data

Categorical → Encoding → OHE

41 & 3200 → OHE ?

18 + 3199 ⇒ CURSE of Dimensionality

19k rows

Resale → Numerical

Encoding

OHE | Label | Target Encoding

Reason?

Maruti → 1
Hyundai → 2
Toyota → 3

Label or Target Encoding

10 - 15 categories → OHE

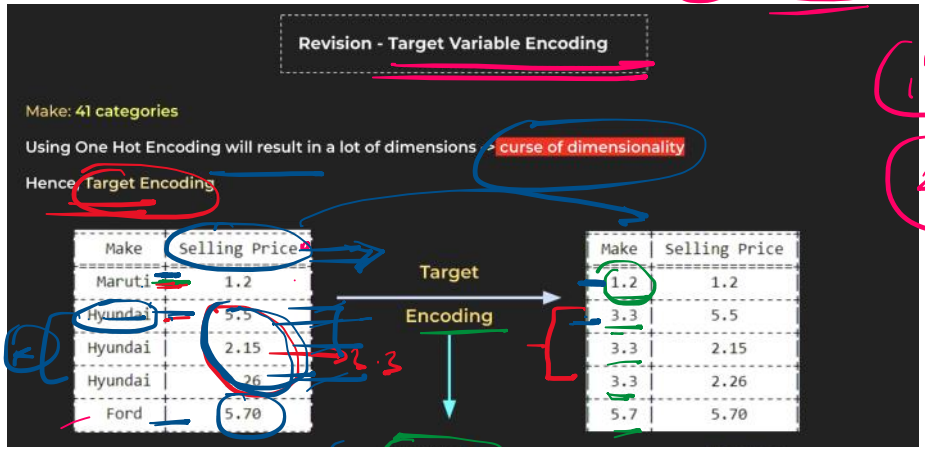
ML → Reduce Training Time ⇒

TARGET ENCODING ⇒

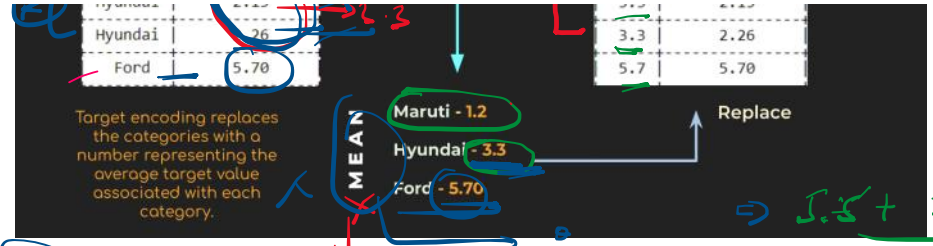
10 → 9 ⇒

PCA

4 3200



Label Encoding



Label Encoding

↓

Random

DATA

① Median → ✓

$\Rightarrow \frac{1.5 + 2.15 + 2.26}{3}$

Lowmoor → Mode

→ Categorical

→ OHE → No. of features

② Encode ⇒ ✗

Label encoding → 1 2 3

③ TARGET encoding ⇒ Replacing

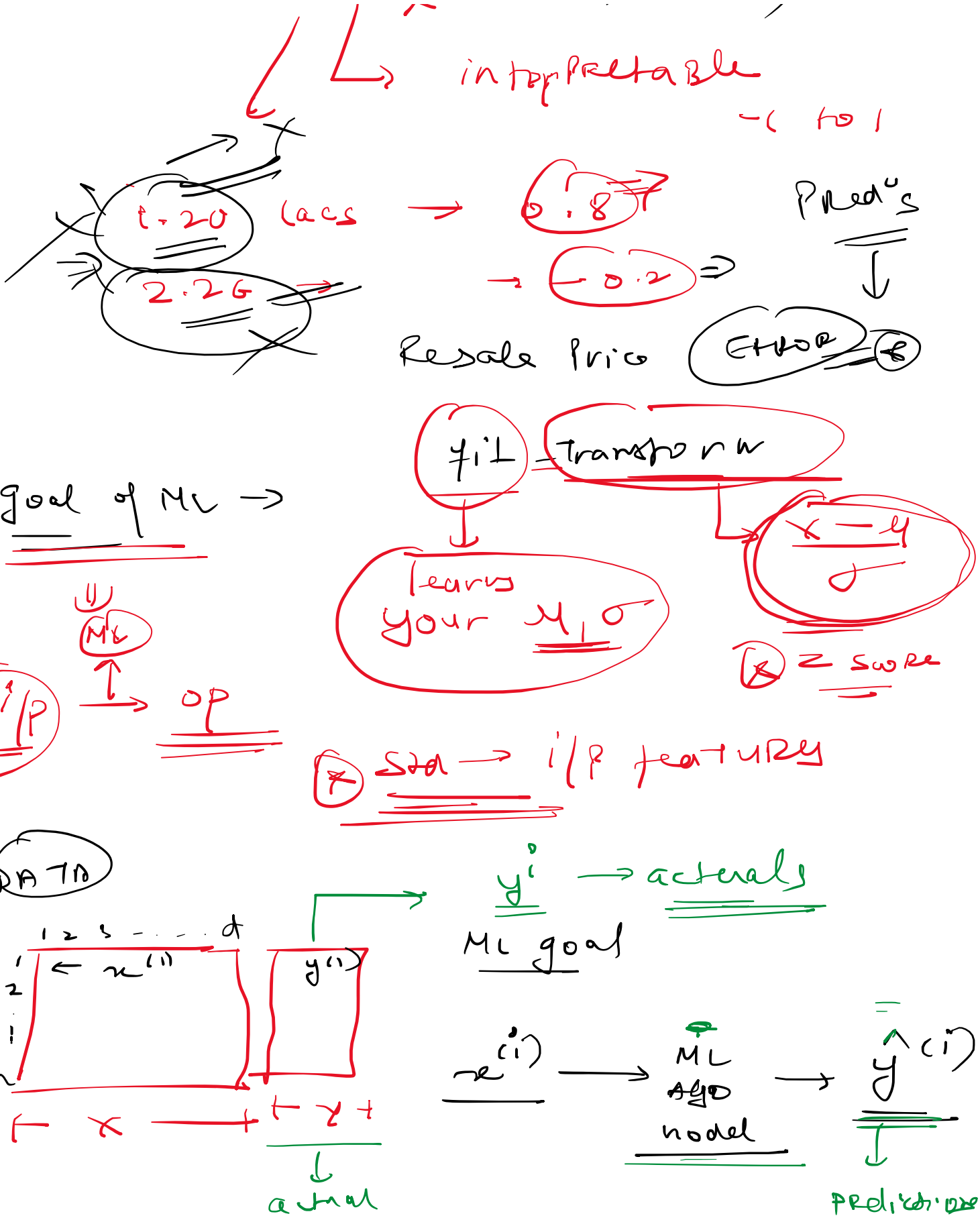
④ Categorical → Encoding ✓ → images

④ Scaling ⇒ Normalization,

⑤ Standardization

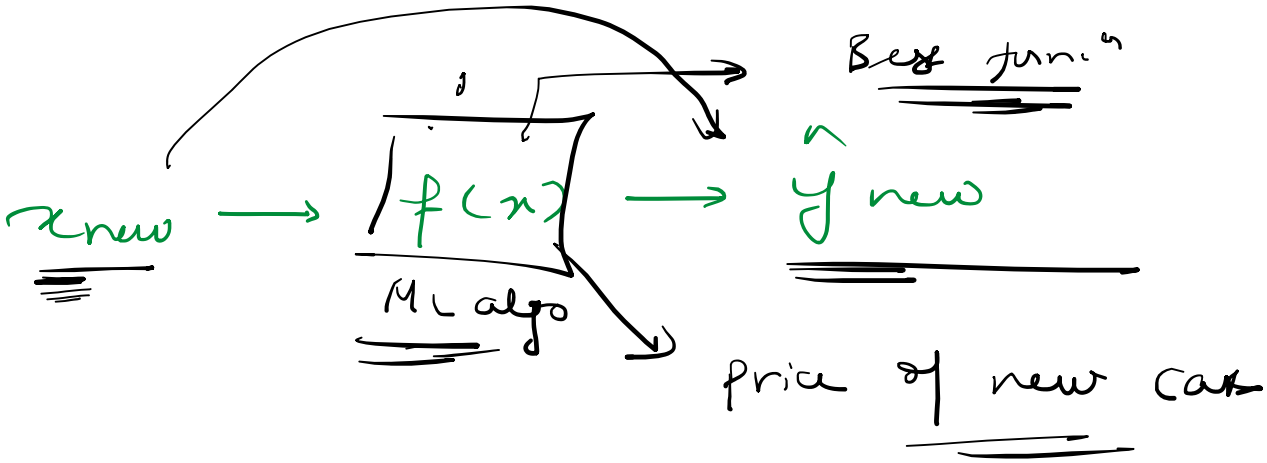
⇒ O/P → Standardization

✗ $\bar{x}$ ✗


interpretable
1.20 lacs
2.26
0.8
0.2
Pred's
Resale Price
Error
goal of ML →
i/p
ML
op
fit
transform
learns your ML o
x — y
z score
std → i/p features
DATA
1 2 3 . . . d
x(i)
y(i)
y → actuals
ML goal
x(i)
ML Algo model
ŷ(i)
x
y
actual
Prediction

ideally $y^{(i)} \approx \hat{y}^{(i)}$

original → Predicted



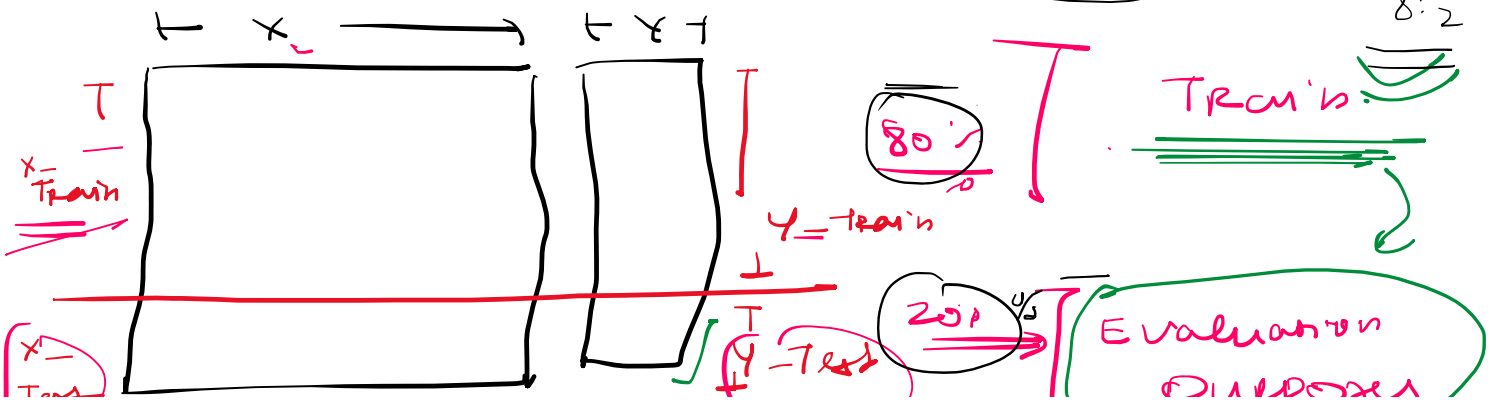
ML

training → on entire data

→ ERRORS ✗

X → Y ⇒ assessment

Splitting of data ⇒

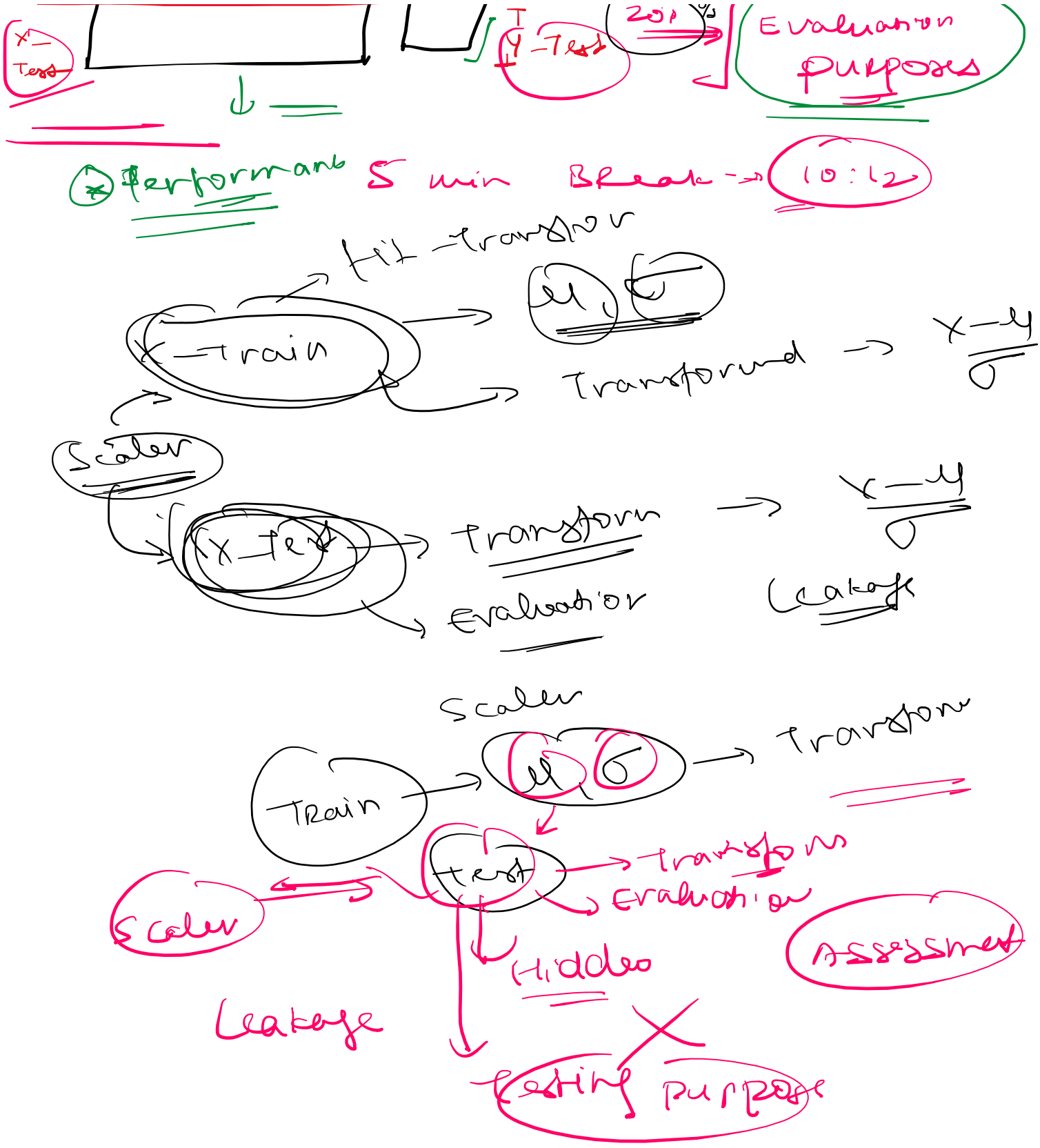


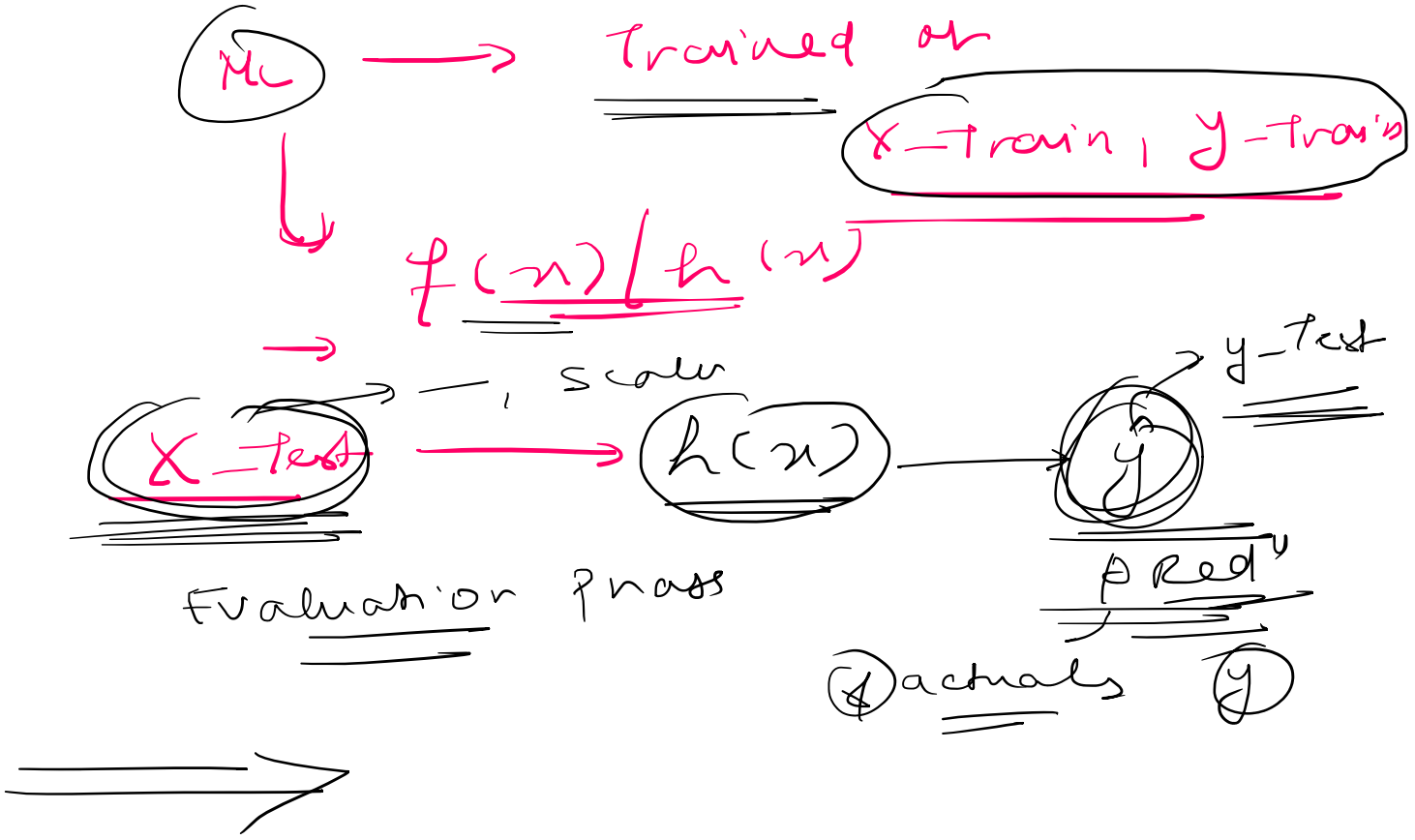
4:1

80 : 20

70 80 60 40 → 7:3 → 6:4

8:2


X_Test
Y_Test
Evaluation purposes
Performance
5 min Break → 10:12
Fit - transform
Train
Transformed → X-y / σ
Scaler
X_Test
transform
Evaluation
X-y / σ
Leakage
Scaler
Train
Test
transform
Evaluation
Scaler
Assessment
Hidden
Leakage
Testing purpose



Linear Regression Intuition ⇒

VARIANTS

~~1~~ UNIVARIATE LR (single variable (1 i/p variable) → Target)

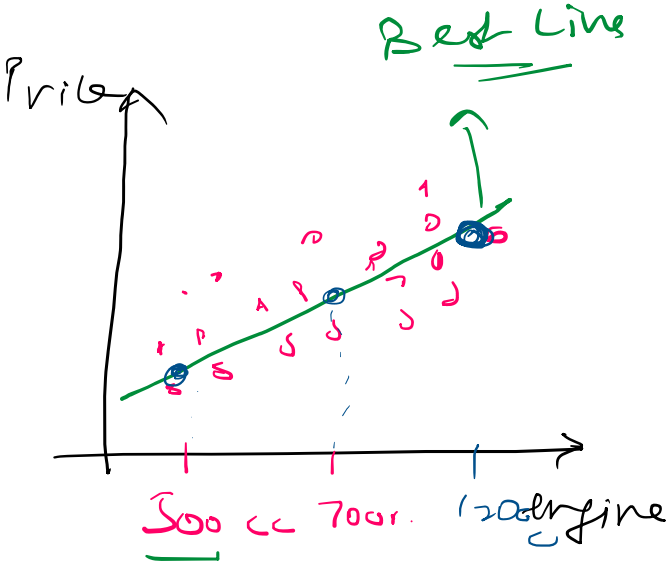
~~2~~ MULTIVARIABLE LR (>1 i/p var) → target

CARS24 ⟹ MULTIVARIATE

Univariate

i/p → o/p

engine → Price



New car (1200 cc)

1200 cc → Price

↳ eq" of line → pred"

x_{new}^{i} → line eq" $h(x)$ → $\hat{y}_{new}^{i}$

St. line ⇒ $y = mx + c$

predicted o/p

data → actual o/p

LR. model

engine

$\hat{y} = w_1 \cdot x + w_0$ ⇒ univariate

$\hat{y}$ → Price

w_1 → weight

w_0 → intercept

Assume ⇒ $\hat{y} = 200 \cdot x + 50000$

engine

w_1 → 200

w_0 → 50000

new i/p → 1200 cc

$\hat{y} = 200 \times 1200 + 50000$

$\hat{y} = 2.9$ lacs (₹)

⇒

1 i/p feature

Line → 2D

2 i/p features ⇒

Price

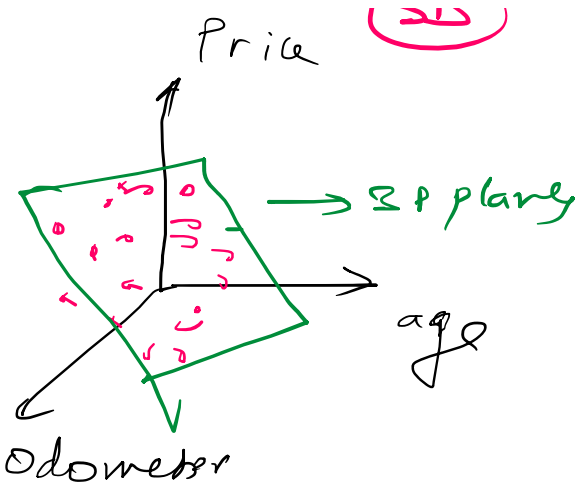
3D

2 i/p features ⇒

Age, odometer (y) Price

Age	odometer	Price

⇒



age ↑ & odometer ↑ ⇒ Price ↓

eqn of plane ⇒

$$y = w_1 x_1 + w_2 x_2 + w_3$$

x_1: age, w_1: weight

x_2: odometer, w_2: weights

$$y = -10000 x_1 - 10 x_2 + 50000$$

x_1: 1 year ⇒ ↓ 10000k

x_2: 1 ⇒ ↓ 10

2 features, 1 feature

2 features

(5D)

d features

$y = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + w_0$

↓ X plot

Algebraic Notation

$y = w^T x + w_0$ (Linear model)

↳ line, plane, hyperplane

1 feature → Line → 2D

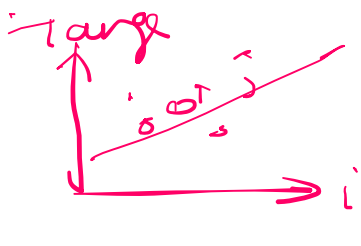
2 features → Plane → 3D

⋮

d dimⁿ → (d+1) dimⁿ hyperplane

$\hat{y} = w^T x + w_0$

↳ Prediction

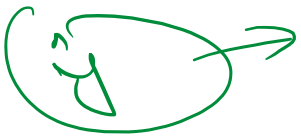


① Best fit line

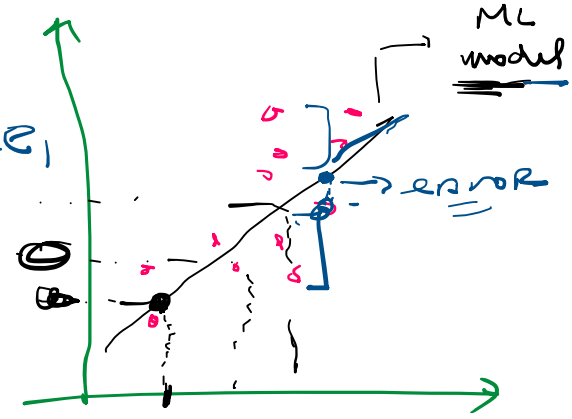
1 feature → 2D line

$\hat{y} = 4.6 \times \text{model} + 6.5$

② ③



④ Evaluation METRICS



$x^{(1)} = y^{(1)} - \hat{y}^{(1)} = \text{ERROR} = e_1$

$x^{(2)} = y^{(2)} - \hat{y}^{(2)} = e_2$

$x_d = e_d$

⇒ Total error = $e_1 + e_2 + \cdots e_d$

$3 + 0 + (-2) + (-4) + 3$

$= 0$ ✗

✓ (BS) value → 1 of them

abs value → 10 + 10

$= 2 \quad |3| + |0| + |-2| + |-4| + 3$

$= \boxed{12} =$

Squaring → 2nd way

$3^2 + 0^2 + (-2)^2 + (-4)^2 + 3^2$

$= \boxed{38}$

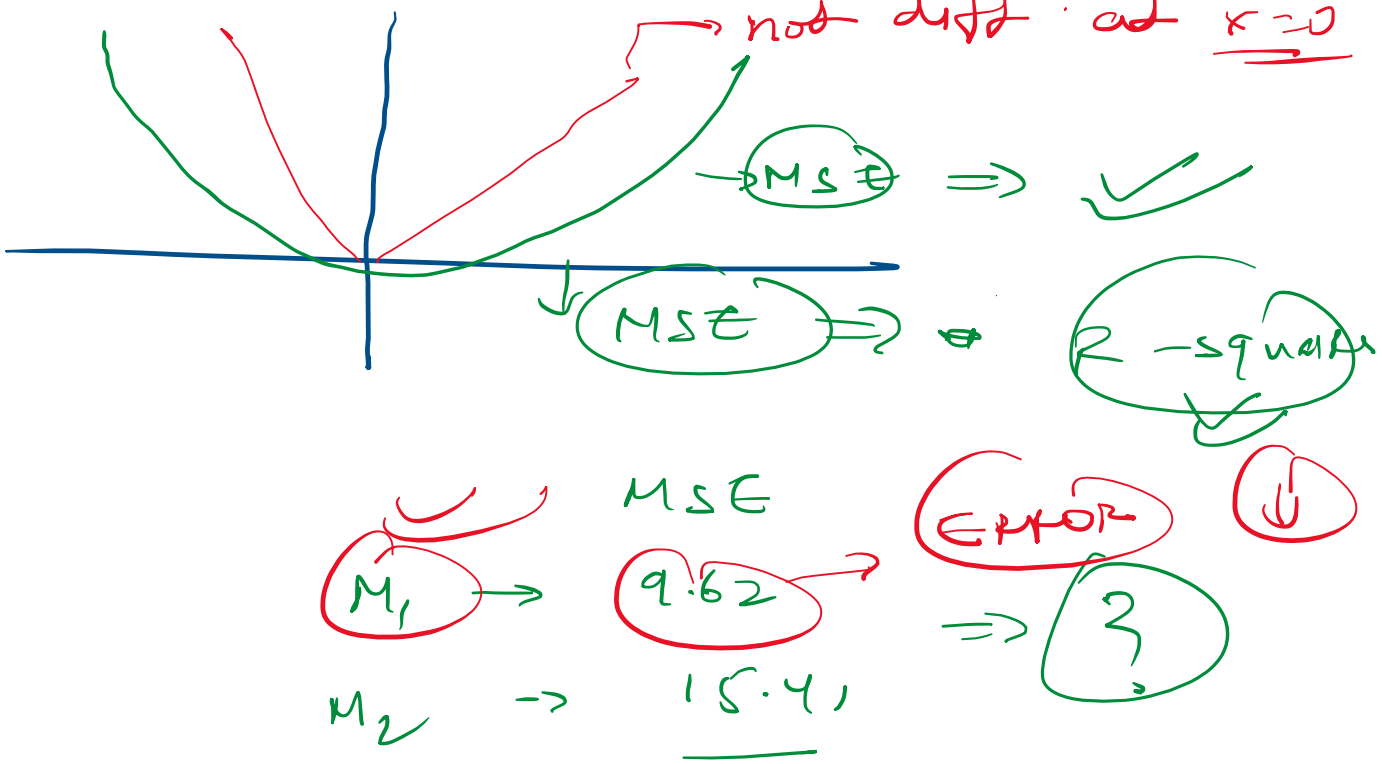
$$\frac{1}{n}\sum_{i=1}^{n} |y^{(i)} - \hat{y}^{(i)}| = \text{MAE}$$

Mean Abs. Error → MAE

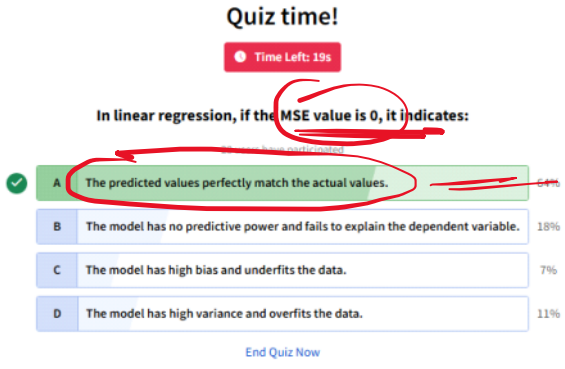
$$\frac{1}{n}\sum_{i=1}^{n} (y^{(i)} - \hat{y}^{(i)})^2 = \text{MSE}$$

Mean Sq. Error → MSE

→ not diff. at $x = 0$



⇒ 1 way of evaluating



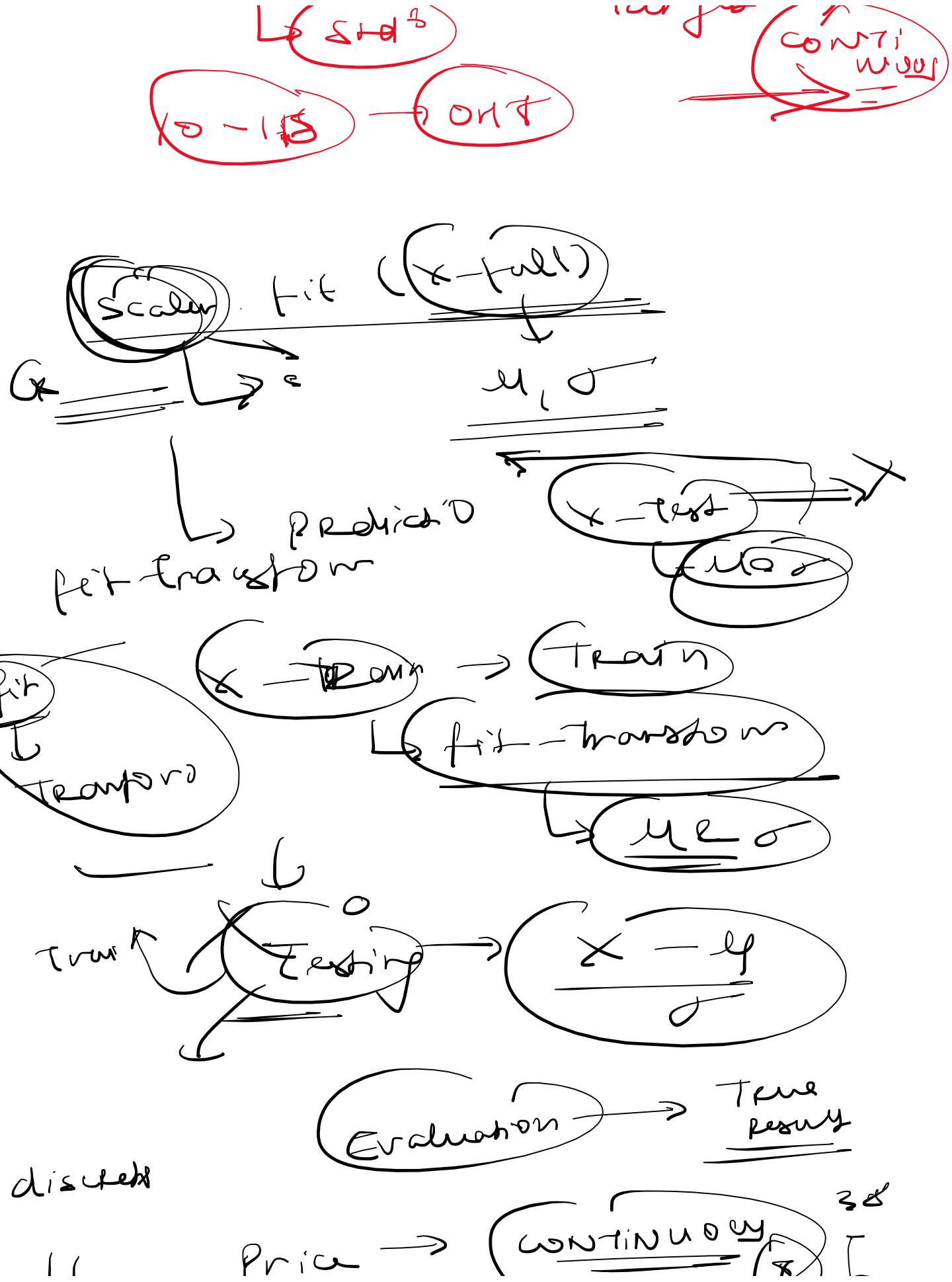
y → actual
$\hat{y}$ → pred^{n}

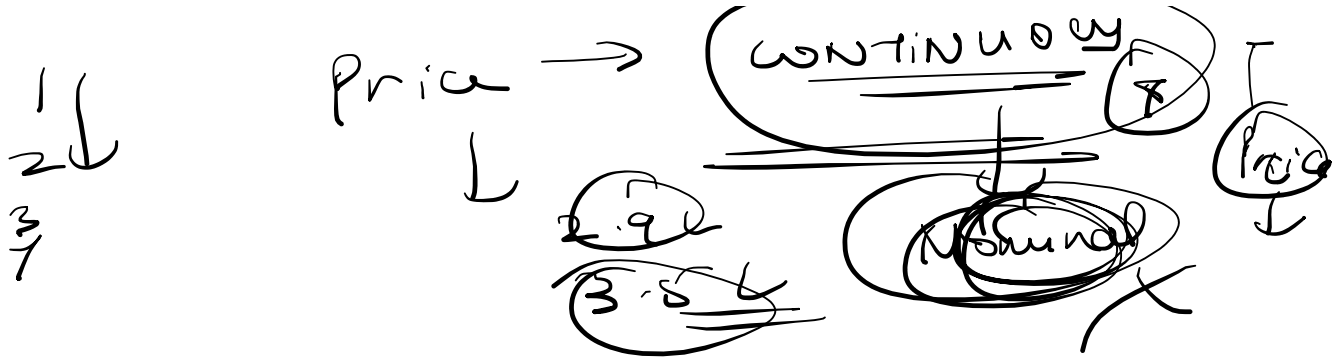
$\hat{y} - y$

$y - \hat{y} = 0$
same

Norm
tabular data
std^{n}

Targets → CONTI


Scaler
fit
x_full
predict
fit_transform
x_test
fit
transform
x_train
fit_transform
Testing
x_y
Evaluation
True results
discrete
Price
continuous


Price
CONTINUOUS
Price
Normal